



Mansoura University
Faculty of Computers and Information
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DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

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COURSE DETAILS

Course Syllabus

Chapter 1: Introduction to DSS

Chapter 2: Decision Making Concepts

Chapter 3: Decision Making Technologies and Environment

Chapter 4: Model Management

Chapter 5: Data Management System and Dialogue Management

Chapter 6: Designing and Developing Decision Support Systems

Chapter 7: Executive Support Systems

Chapter 8: Automated Decision Systems and Expert Systems

Chapter 9: Knowledge Management and Collaborative Systems

Chapter 10: DSS Intelligent Systems & Tools and Techniques

Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

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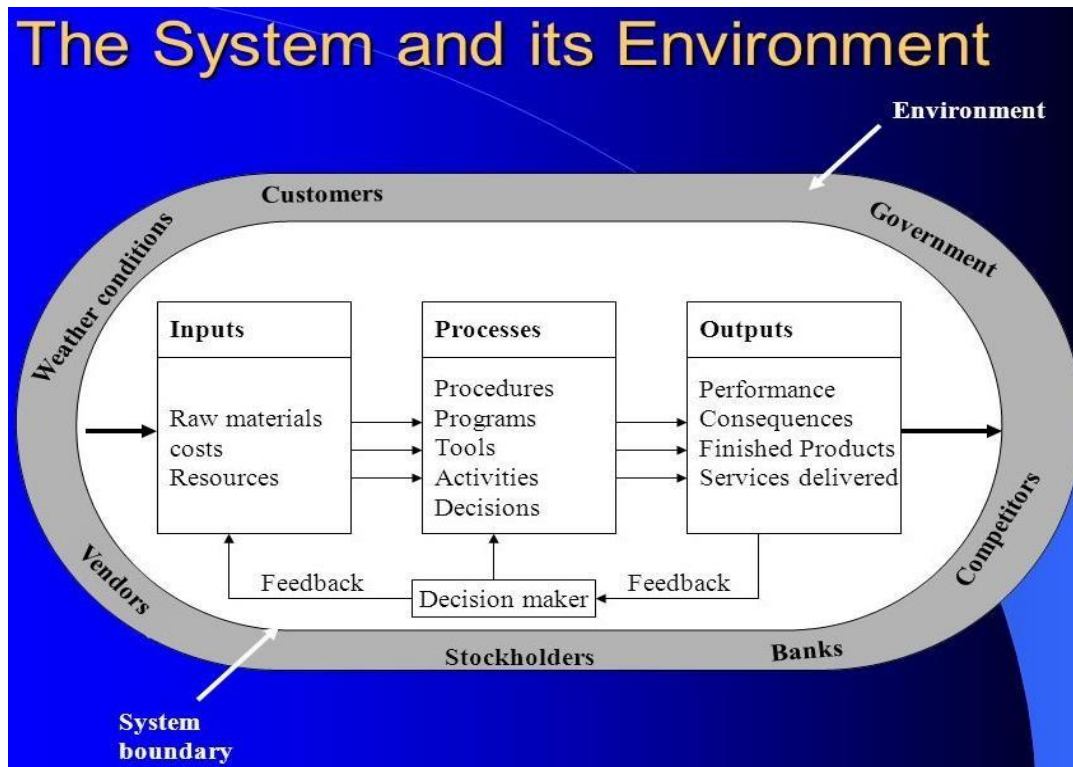
Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

CHAPTER I CONTENTS

- Systems
- Information
- Information systems (IS)
- Management information systems (MIS)
- Efficient and effective systems
- Components of IS
- Types of IS
- DSS evolution

SYSTEMS

A System: an interrelated set of components including people, activities, technology, and procedures that are designed or intended to achieve a predefined purpose.



MANAGERIAL DECISION MAKING AND INFORMATION SYSTEMS

- **Management** is a process by which organizational goals are achieved through the use of resources



- **Resources:** *Inputs*
- **Goal** Attainment: *Output*
- Measuring **Success**:

$$\text{Productivity} = \text{Outputs} / \text{Inputs}$$

Management is decision making

The **manager** is a decision maker

Now fast changing, complex environment

Trial-and-error: not a great approach

INFORMATION SYSTEMS AND ITS CONCEPTS (I OF 2)

- **Decision** is a choice made between alternative courses of action in a situation of uncertainty.
- **Knowledge** awareness or familiarity gained by experience of a fact or situation. Knowledge comprises things that are held to be true in each context and that drive us to action if there were no impediments.
- **Information**, defined as processed data is an increment in knowledge: it contributes to the general framework of concepts and facts that we know.
- **Data** are only the raw facts, the material for obtaining information. Information systems use data stored in computer databases to provide needed information.

INFORMATION SYSTEMS AND ITS CONCEPTS (2 OF 2)

- **A database** is an organized collection of interrelated data reflecting a major aspect of a firm's activities.
- **Downsizing** In information systems, transferring some or all the organization computing from centralized processing on mainframes or minicomputers to systems built around networked microcomputers (often in a client/server configuration).

INFORMATION AND ITS ROLE IN BUSINESS

Information systems **capture data** from the organization (internal data) and its environment (external data).



They **store** the database items over an extensive period.



When specific information is needed, the appropriate data items are **manipulated** as necessary, and the user receives the resulting information.

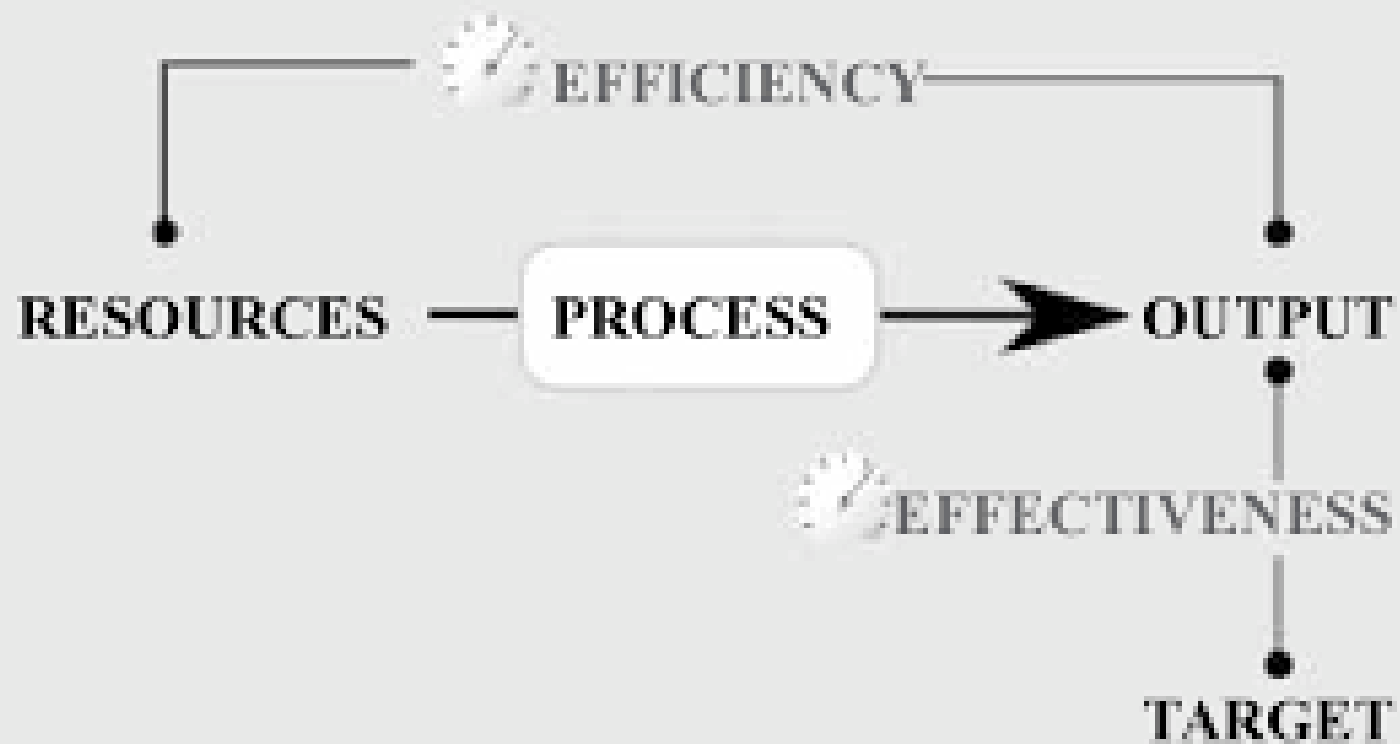


Depending on the type of information system, the information **output** may take the form a **query response, decision outcome, expert-system advice, transaction document, or a report**

QUALITY INFORMATION NEEDS TO POSSESS SEVERAL ATTRIBUTES

- **Timely:** Available when needed and not outdated when made available
 - **Complete:** Includes all the user needs to know about the situation where the information will be used
 - **Concise:** Does not include elements unneeded by the user
 - **Relevant:** Has direct bearing on the situation
 - **Precise:** Offers quantitative information with a degree of exactness appropriate to the underlying data
 - **Form:** The level of detail, tabular versus graphic display, and quantitative versus qualitative form are selected in accordance with the situation
- Internal and External Information

EFFECTIVENESS AND EFFICIENCY



Effectiveness (Right way)

COMPONENTS OF INFORMATION SYSTEMS

- **Information systems consist of the following general components:**
 - Hardware
 - Software
 - Databases
 - Human resources
 - Procedures

TYPES OF INFORMATION SYSTEMS

- Organizations employ several types of information systems. These include:

- 1) Transaction Processing Systems (TPS)
- 2) Management Reporting Systems (MRS)
- 3) Decision Support Systems (DSS)
- 4) Executive Information Systems (EIS)
- 5) Office Information Systems (OIS)
- 6) Professional Support Systems
- 7) Expert systems (ES)

TRANSACTION PROCESSING SYSTEMS

- **A Transaction** is an elementary **activity** conducted during business operations.
- **TPS** may work either in **batch mode**, processing accumulated transactions at a single time later on, or in **on-line** mode, processing incoming transactions immediately.
- Transaction processing systems **today** generally work in **on-line mode** by immediately processing a firm's business transactions.

MANAGEMENT REPORTING SYSTEMS

- The objective of management reporting systems is to provide routine information to managers.
- Managers receive performance **reports** within their specific areas of responsibility.
- Generally, these reports provide internal information rather than spanning corporate boundaries.
- They report on the **past** and the **present**, rather than projecting the future.

DECISION SUPPORT SYSTEMS

- Decision support systems directly support a **decision-making session**.
- These systems facilitate a **dialog between the user, who is considering alternative problem solutions**, and the system that provides **built-in models** and access to databases.
- The DSS databases are often extracted from the general databases of the enterprise or from external databases.

EXECUTIVE INFORMATION SYSTEMS

- Executive information systems support **top managers** with conveniently displayed **summarized information**, customized for them. They make a variety of internal and external information readily available in a highly summarized and convenient form. **EIS are used to:**
 - Monitor the performance of the organization.
 - Assess the business environment.
 - Develop strategic directions for the company's future.

OFFICE INFORMATION SYSTEMS

- The main objective of OIS is to facilitate **communication** between the members of an organization and between the organization and its environment. **OIS are used to:**
 - Help manage documents represented in an electronic format.
 - Handle messages, such as electronic mail, facsimile, and voice mail
 - Facilitate teleconferencing and electronic meetings.
 - Facilitate the use of the Internet for communication and access to information.
 - Facilitate the use of task-oriented teams using groupware.

EXPERT SYSTEMS IN INFORMATION SYSTEMS

- **Expert systems** are system that employs **knowledge** about its application domain and uses an **inferencing** (reason) procedure to solve problems that would otherwise require human competence or expertise

EVOLUTION OF DECISION SUPPORT SYSTEMS

- **mid-1950s** Transaction Processing Systems (TPS)
- **1960s** MIS
- **1970s** Office Automation Systems
DSS
- **1980s** DSS Expanded
Commercial applications of expert systems
Executive Information Systems
- **1990s** Group Support Systems
Neural Computing
Integrated, hybrid computer systems

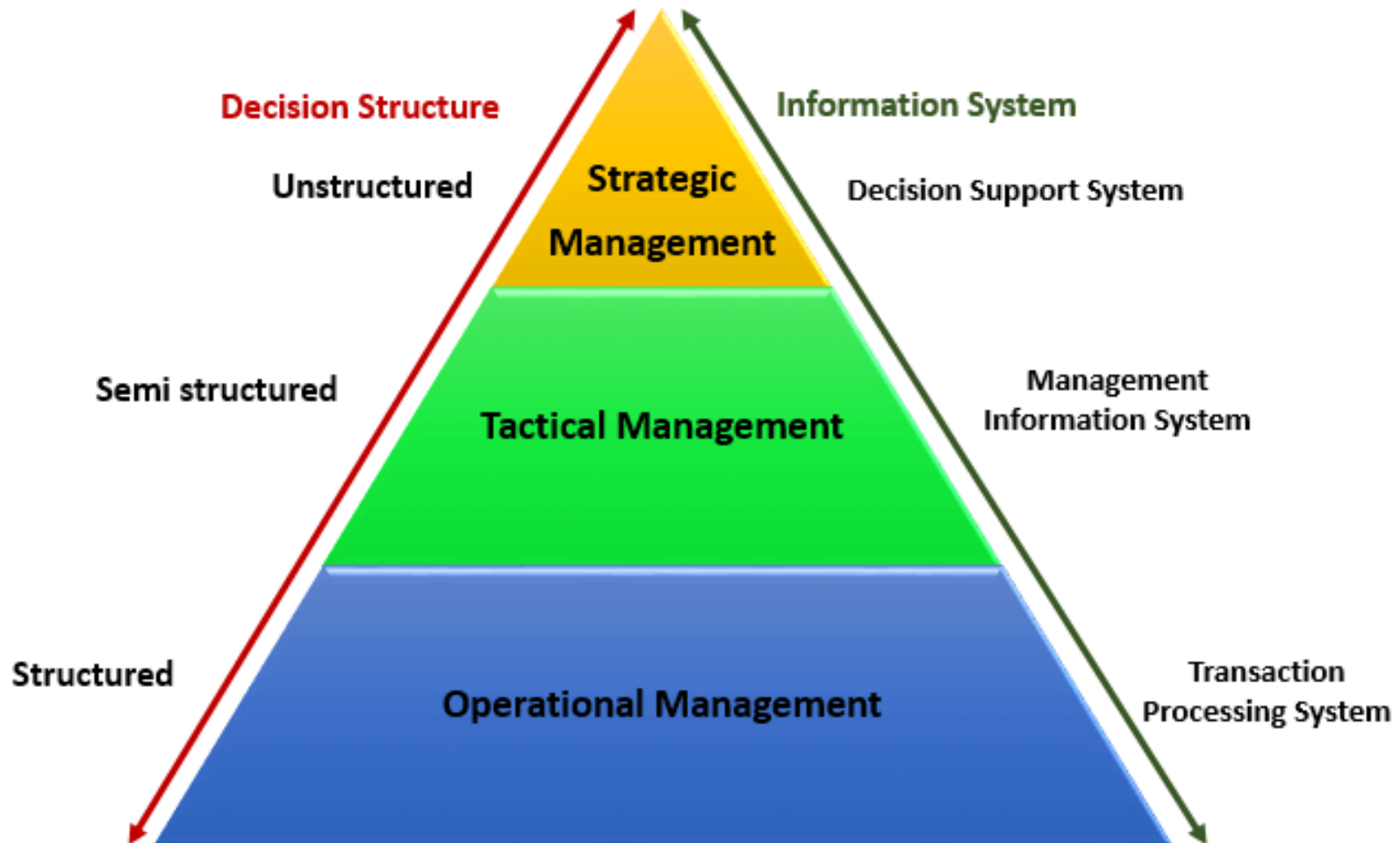
MANAGERS AND DECISION MAKING: WHY COMPUTERIZED SUPPORT?

- Speedy computations
- Overcome cognitive limits in processing and storage
- Cognitive limits may restrict an individual's problem-solving capability
- Cost reduction
- Technical support
- Quality support
- Competitive edge
- The *MANAGERS* are always responsible for decision making

MANAGERS AS INFORMATION SYSTEM USERS

- Managers exist at various **managerial levels** and within various **business areas** of the firm
- The 3 primary management levels are:
 1. **Strategic** planning level
 2. **Management** control level
 3. **Operational** control level

DSS AND MANAGEMENT



THE FUNDAMENTAL FUNCTIONS OF MANAGEMENT INCLUDE:

- **Planning:** establishing goals and selecting the actions needed to achieve them over a specific period of time.
- **Organizing:** This process of establishing worker relationships allows workers to work together to achieve their organizational goals.
- **Leadership** including the people in the organization to contribute to its goals.
- **Staffing:** Recruiting and selecting employees for positions within the company (within teams and departments).
- **Controlling:** measuring performance against the planned objectives and initiating corrective action, if needed.

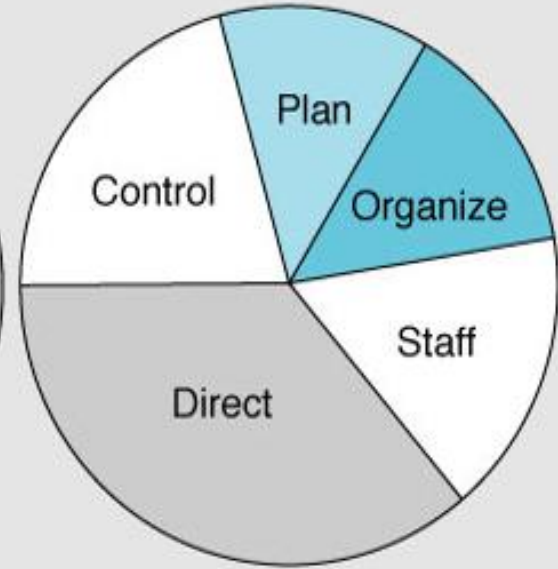
Strategic planning level



Management control level



Operational control level



THE PROCESS OF DECISION MAKING

- **Defining the problem** (a decision situation that may deal with some difficulty or with an opportunity)
- **Constructing a model** that describes the real-world problem
- Identifying **possible solutions** to the modeled problem and evaluating the solutions
- **Comparing, choosing, and recommending** a potential solution to the problem

DECISION MAKING

Process of choosing amongst alternative courses of action for the purpose of attaining a goal or goals.

- **The four phases of the decision process are:**
 - Intelligence
 - Design
 - Choice
 - Implementation



Thanks
Any questions